Preprocessing of the experimental data reported in *Neglect-Zero effects in the interpretation of quantifiers and disjunction*

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Preprocessing

Import data from .csv

```
rm(list=ls())
source("sdTrimBugFixed.R")

plots_dir<- file.path(getwd(),"plots")
if(!dir.exists(plots_dir)) dir.create(plots_dir, recursive = TRUE, showWarnings = FALSE)

subj_info <- read.csv(file = file.path("..", "data", "hohenentringen_subj_info.csv"))
data <- read.csv(file = file.path("..", "data", "hohenentringen.csv"))
```

Some basic information about data

Show how many pps

```
sum(xtabs(~id, data=data)/205)
```

```
## [1] 80
```

Show how many trials each pp has done

```
xtabs(~id, data)
```

```
## id
## 65b3fa983e373 65b3ff9b03566 65b40096a9376 65b4042455fcd 65b43998c5a4d
##           205           205           205           205           205
## 65b447741132b 65b4501a1f627 65b4543bd545c 65b4580ac0580 65b45de93a1f4
##           205           205           205           205           205
## 65b49cbce7899 65b4a0c89f9c7 65b4a1e9b526d 65b4aeee17d37 65b4bb92f01ff
##           205           205           205           205           205
## 65b4bd0c79a85 65b4bf56be61c 65b4c4430e966 65b4c545bd8bc 65b4c69d38697
##           205           205           205           205           205
## 65b4c6da204cb 65b4c8181c444 65b4c87d76174 65b4c92488912 65b4c94d528a0
##           205           205           205           205           205
## 65b4ca6ada848 65b4cae3cb20b 65b4ccad59798 65b4cd1f2e66e 65b4cec6000bc
##           205           205           205           205           205
## 65b4d5aa97e37 65b4d9cbc2f15 65b4e47754770 65b4e7d25d24f 65b4ea615241a
##           205           205           205           205           205
```

```
## 65b4eea24d8c5 65b4ef2037902 65b4ef5e1df62 65b4ef8354020 65b4efcd033ac
##           205           205           205           205           205
## 65b4effcf247c 65b4f276bcd27 65b4f4d723d7c 65b4f665707e0 65b4ffdbcf468
##           205           205           205           205           205
## 65b50327bfb3c 65b5061044616 65b50a02eab8e 65b50a5c1489d 65b50ab8e8d5b
##           205           205           205           205           205
## 65b5103a0ddaf 65b510a106b2c 65b5145c7055e 65b516b8e7979 65b516f90b6c5
##           205           205           205           205           205
## 65b519ca4c4f9 65b51af69b6e1 65b51b80c4e3d 65b520292dacf 65b520b51abd4
##           205           205           205           205           205
## 65b523efe0c91 65b5255086520 65b5280a4ec32 65b52c474ecda 65b533a573cde
##           205           205           205           205           205
## 65b534834b712 65b53b1fd6c3b 65b53d8fda9f4 65b54007bd413 65b54899d969d
##           205           205           205           205           205
## 65b54eb781604 65b5576b5dbb2 65b55beec50f1 65b55eff62725 65b561b3594bf
##           205           205           205           205           205
## 65b566fec0c2f 65b58b1700b54 65b64a18444c5 65b65aead7fff 65b673260eb94
##           205           205           205           205           205
```

Show more stuff

```
round(xtabs(~condition,data))
xtabs(~id+list, data=data)
xtabs(~item+condition, data=data)
```

Prepare pp exclusion

```
exclude <- NULL
```

Show missing reading times

```
xtabs(~data$time.out=="1",data)
```

```
## data$time.out == "1"
## FALSE  TRUE
## 16399     1
```

Mark all time out data as NA

```
data$pictureRT[which(data$time.out=="1")] <- NA
xtabs(~is.na(data$pictureRT),data)
```

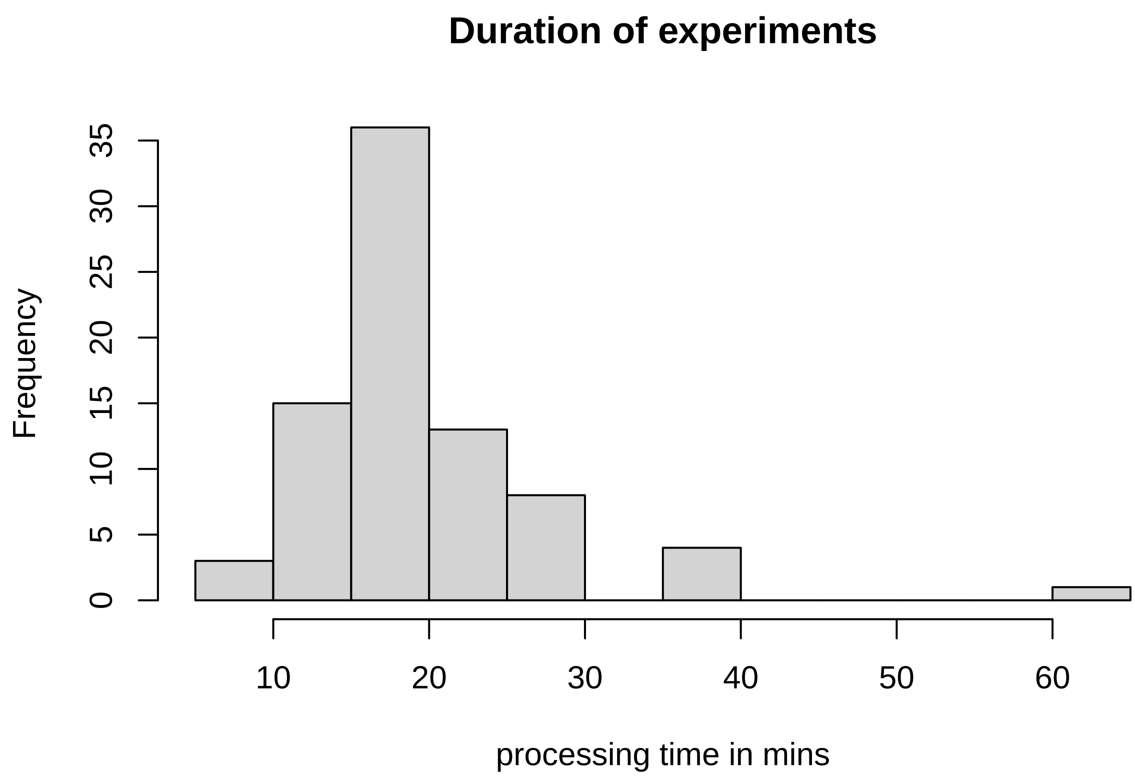
```
## is.na(data$pictureRT)
## FALSE  TRUE
## 16399     1
```

Remove trials with NA values in pic rt

```
data <- subset(data, !is.na(data$pictureRT))
# nrow(subset(data, is.na(data$id)))
# data <- subset(data, !is.na(data$id))
```

Show average duration for experiments

```
hist(subj_info$time_in_minutes, main = "Duration of experiments", xlab = "processing time in mins")
```




```
mean(subj_info$time_in_minutes) # N = 19.7
```

```
## [1] 19.6235
```

```
sd(subj_info$time_in_minutes)   # SD = 7.8
```

```
## [1] 7.599848
```

Show pps more than three SD away from mean and add them to exclude list

```
distance <- 3
x <- mean(subj_info$time_in_minutes) + distance * sd(subj_info$time_in_minutes)
y <- mean(subj_info$time_in_minutes) - distance * sd(subj_info$time_in_minutes)
filter(subj_info, time_in_minutes < y | time_in_minutes > x)
```

```
##              id missed_break_one missed_break_two missed_break_three
## 1 65b50a02eab8e            False            False              False
##   time_in_minutes language asses age gender
## 1        60.33597  deutsch   Yes  23   Male
##                                                              comments fluent
## 1 Zweideutige Interpretation der Antwortmöglichkeiten bzw. Fragen.    Yes
```

```
(exclude <- append(exclude, filter(subj_info, time_in_minutes < y | time_in_minutes > x)$id))
```

```
## [1] "65b50a02eab8e"
```

Show missing reading times

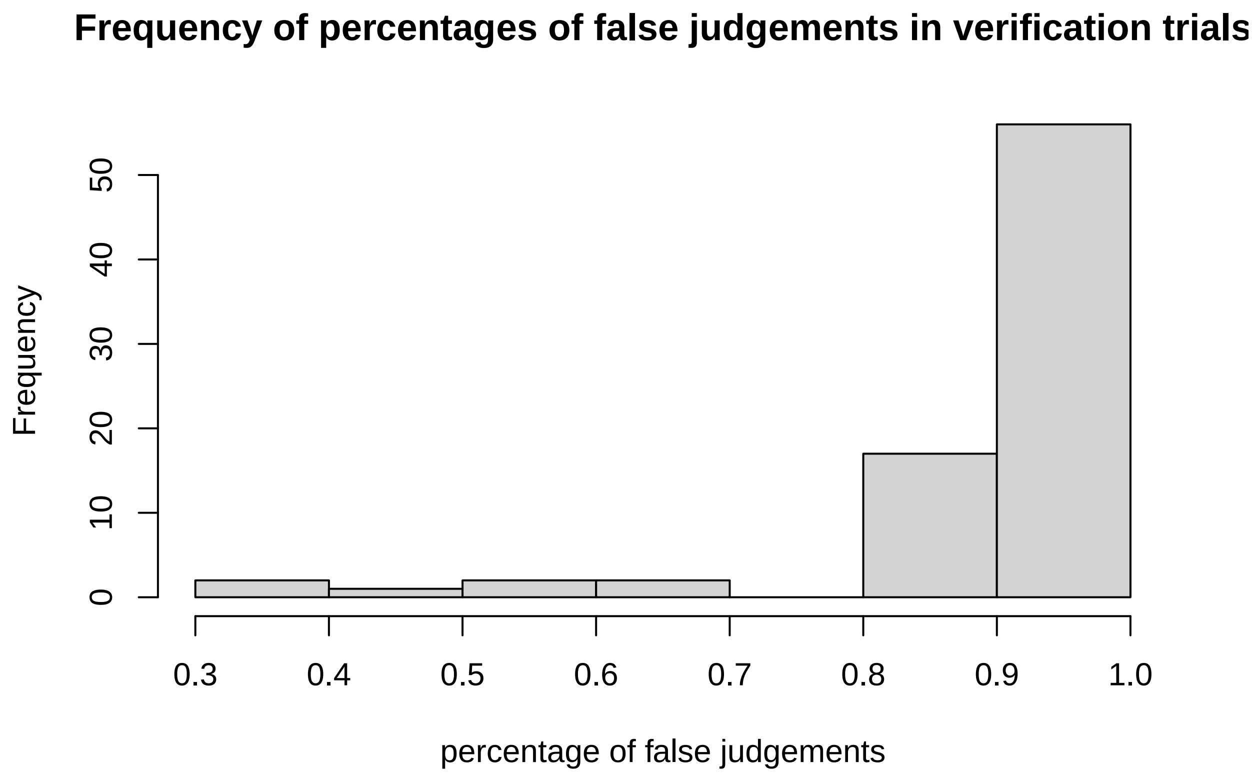
```r
sort(tapply(data$ordered_responses, c(data$response_order), first))
```

```
##                                                           2
## "[['ja, stimmt'], ['komische Frage'], ['nein, stimmt nicht']]"
##                                                           1
## "[['ja, stimmt'], ['nein, stimmt nicht'], ['komische Frage']]"
##                                                           5
## "[['komische Frage'], ['ja, stimmt'], ['nein, stimmt nicht']]"
##                                                           6
## "[['komische Frage'], ['nein, stimmt nicht'], ['ja, stimmt']]"
##                                                           3
## "[['nein, stimmt nicht'], ['ja, stimmt'], ['komische Frage']]"
##                                                           4
## "[['nein, stimmt nicht'], ['komische Frage'], ['ja, stimmt']]"
```

```r
#data$given_response <- ""
data$given_response <- case_when(data$response==37 ~ case_when(data$response_order %in% c(1,2)
                                                                ~ "ja, stimmt",
                                                                data$response_order %in% c(3,4)
                                                                ~ "nein, stimmt nicht",
                                                                data$response_order %in% c(5,6)
                                                                ~ "komische Frage"),
                                 data$response==40 ~ case_when(data$response_order %in% c(5,3)
                                                                ~ "ja, stimmt",
                                                                data$response_order %in% c(1,6)
                                                                ~ "nein, stimmt nicht",
                                                                data$response_order %in% c(2,4)
                                                                ~ "komische Frage"),
                                 data$response==39 ~ case_when(data$response_order %in% c(4,6)
                                                                ~ "ja, stimmt",
                                                                data$response_order %in% c(2,5)
                                                                ~ "nein, stimmt nicht",
                                                                data$response_order %in% c(1,3)
                                                                ~ "komische Frage"))
data$accuracy <- NULL
data$accuracy <- ifelse(tolower(data$given_response) == tolower(data$expected_response), 1, 0)
check_response_coding<-data[,c("response","ordered_responses","given_response","expected_response","accu
```

Histogram of accuracy In (real) fillers (items 1000+) and/or 'homogeneity' items correctly

```r
data$item <- as.numeric(data$item)
#attention_controls <- data[data$item >= 1000,]#||
attention_controls <- data[(data$item>= 200 & data$item < 300 & data$condition %in% c(1,2,4)),]


control_accuracy<-aggregate(attention_controls$accuracy,list(attention_controls$id),mean,na.rm=TRUE)
names(control_accuracy)<- c("id","accuracy")

hist(control_accuracy$accuracy, main = "Frequency of percentages of false judgements in verification tri
```



Show how many pps are excluded under a given cut-off of 80%

```
sum(control_accuracy$accuracy>=.8)
```

```
## [1] 73
```

```
sum(control_accuracy$accuracy<.8)
```

```
## [1] 7
```

```
(exclude <- append(exclude, control_accuracy[control_accuracy$accuracy<.8,]$id))
```

```
## [1] "65b50a02eab8e" "65b4501a1f627" "65b49cbce7899" "65b4ea615241a"
## [5] "65b4f276bcd27" "65b5145c7055e" "65b520b51abd4" "65b54007bd413"
```

Identify and exclude trials with extremely long reading times

```
#melt_data_read <- subset(melt_data, roi!= "pictureRT" & !is.na(melt_data$rt))
#with(melt_data_read, plot(rt, col=participant_id))
data$read_time<-as.numeric(str_sub(data$read_time,2,-2))
cutoff_ex_rt <- 10000
data$extreme_read_time <- ifelse(data$read_time>cutoff_ex_rt,1,0)
aggregate(data$extreme_read_time,list(data$id),mean,na.rm=T)
```

```
##          Group.1           x
## 1  65b3fa983e373 0.000000000
## 2  65b3ff9b03566 0.000000000
```

```
## 3  65b40096a9376 0.000000000
## 4  65b4042455fcd 0.000000000
## 5  65b43998c5a4d 0.039024390
## 6  65b447741132b 0.000000000
## 7  65b4501a1f627 0.014634146
## 8  65b4543bd545c 0.000000000
## 9  65b4580ac0580 0.000000000
## 10 65b45de93a1f4 0.000000000
## 11 65b49cbce7899 0.000000000
## 12 65b4a0c89f9c7 0.000000000
## 13 65b4a1e9b526d 0.000000000
## 14 65b4aeee17d37 0.000000000
## 15 65b4bb92f01ff 0.000000000
## 16 65b4bd0c79a85 0.000000000
## 17 65b4bf56be61c 0.000000000
## 18 65b4c4430e966 0.000000000
## 19 65b4c545bd8bc 0.000000000
## 20 65b4c69d38697 0.000000000
## 21 65b4c6da204cb 0.000000000
## 22 65b4c8181c444 0.000000000
## 23 65b4c87d76174 0.000000000
## 24 65b4c92488912 0.000000000
## 25 65b4c94d528a0 0.000000000
## 26 65b4ca6ada848 0.000000000
## 27 65b4cae3cb20b 0.000000000
## 28 65b4ccad59798 0.000000000
## 29 65b4cd1f2e66e 0.000000000
## 30 65b4cec6000bc 0.000000000
## 31 65b4d5aa97e37 0.034146341
## 32 65b4d9cbc2f15 0.000000000
## 33 65b4e47754770 0.000000000
## 34 65b4e7d25d24f 0.000000000
## 35 65b4ea615241a 0.000000000
## 36 65b4eea24d8c5 0.000000000
## 37 65b4ef2037902 0.024390244
## 38 65b4ef5e1df62 0.000000000
## 39 65b4ef8354020 0.048780488
## 40 65b4efcd033ac 0.000000000
## 41 65b4effcf247c 0.000000000
## 42 65b4f276bcd27 0.000000000
## 43 65b4f4d723d7c 0.000000000
## 44 65b4f665707e0 0.009756098
## 45 65b4ffdbcf468 0.000000000
## 46 65b50327bfb3c 0.000000000
## 47 65b5061044616 0.000000000
## 48 65b50a02eab8e 0.029268293
## 49 65b50a5c1489d 0.000000000
## 50 65b50ab8e8d5b 0.000000000
## 51 65b5103a0ddaf 0.000000000
## 52 65b510a106b2c 0.000000000
## 53 65b5145c7055e 0.000000000
## 54 65b516b8e7979 0.000000000
## 55 65b516f90b6c5 0.000000000
## 56 65b519ca4c4f9 0.000000000
```

```
## 57 65b51af69b6e1 0.014634146
## 58 65b51b80c4e3d 0.000000000
## 59 65b520292dacf 0.000000000
## 60 65b520b51abd4 0.004878049
## 61 65b523efe0c91 0.000000000
## 62 65b5255086520 0.000000000
## 63 65b5280a4ec32 0.000000000
## 64 65b52c474ecda 0.000000000
## 65 65b533a573cde 0.004878049
## 66 65b534834b712 0.000000000
## 67 65b53b1fd6c3b 0.000000000
## 68 65b53d8fda9f4 0.000000000
## 69 65b54007bd413 0.000000000
## 70 65b54899d969d 0.009756098
## 71 65b54eb781604 0.000000000
## 72 65b5576b5dbb2 0.000000000
## 73 65b55beec50f1 0.000000000
## 74 65b55eff62725 0.000000000
## 75 65b561b3594bf 0.000000000
## 76 65b566fec0c2f 0.000000000
## 77 65b58b1700b54 0.000000000
## 78 65b64a18444c5 0.000000000
## 79 65b65aead7fff 0.004878049
## 80 65b673260eb94 0.000000000
```

```
#Sine no pp has extreme reading times in more than 5% of the trials and they are distributed evenly acr
#TODO: exclude >5%
filtered_data <- subset(data, data$read_time<=cutoff_ex_rt)
```

Remove excluded pps

```
filtered_data <- subset(filtered_data, !(id %in% exclude))
```

Check how many pps remained

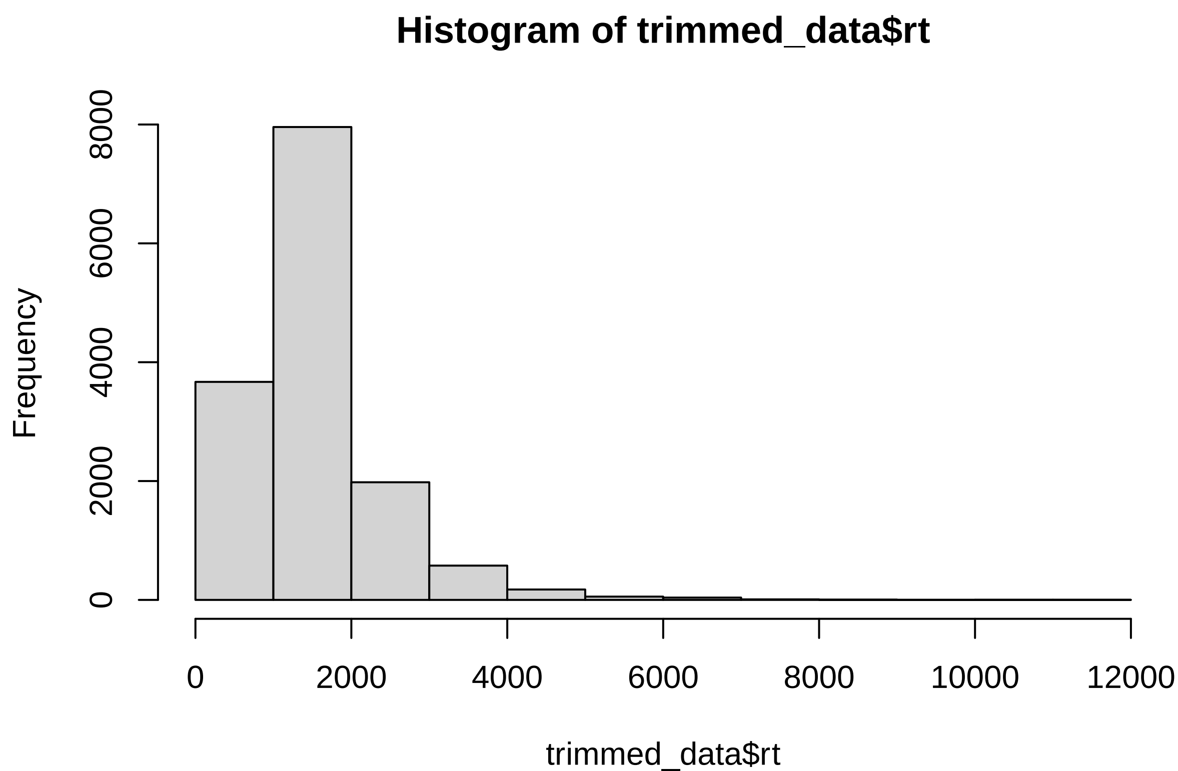
```
length(unique(filtered_data$id))
```

```
## [1] 72
```

Use trimr to trim picture rt

```
trimmed_data <- filtered_data

trimmed_data$cond <- trimmed_data$condition
trimmed_data$acc <- trimmed_data$accuracy

trimmed_data$pictureRT <- as.numeric(trimmed_data$pictureRT)
trimmed_data$rt = trimmed_data$pictureRT
trimmed_data$participant <- trimmed_data$id
trimmed_data$condition <- as.factor(paste(trimmed_data$experiment))
trimmed_data$accuracy <- 1 #(accuracy only because trimr needs this colmun name to function)


trimmed_data <- sdTrimBugFixed(data = trimmed_data, minRT = 200, sd = 3,
```

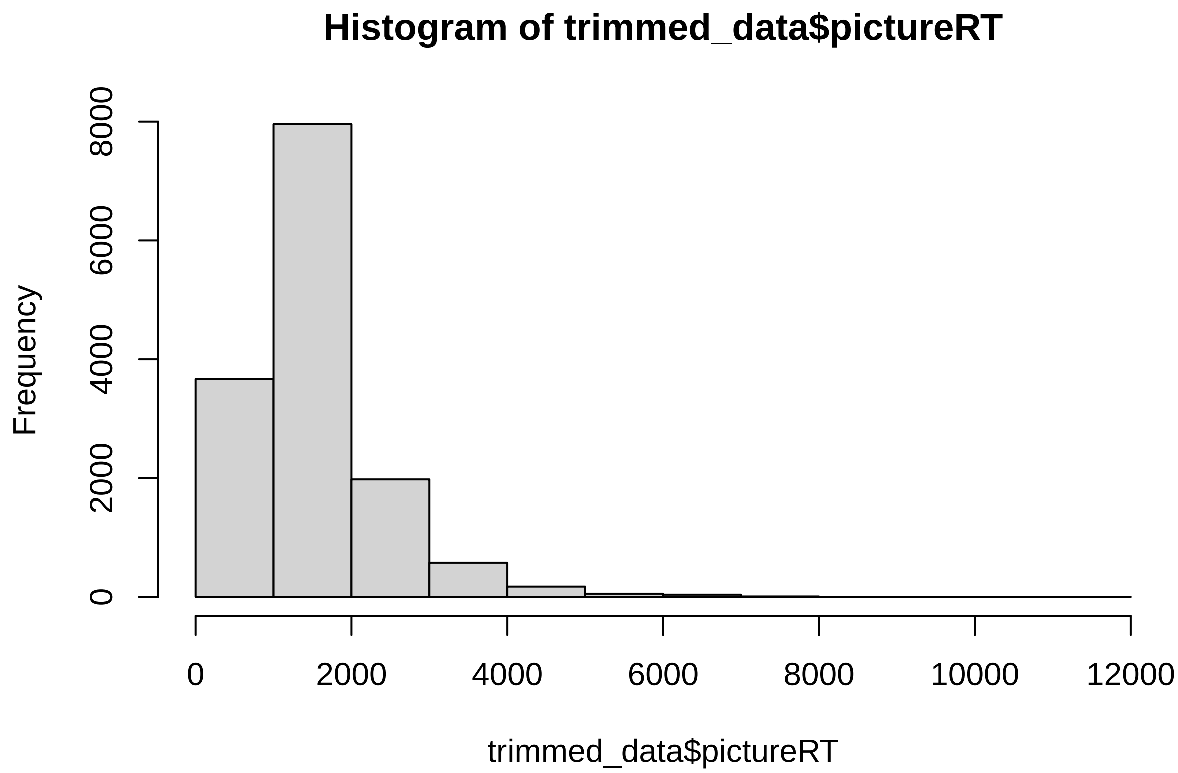
```r
                              perCondition = TRUE, perParticipant = TRUE,
                              omitErrors = FALSE, returnType = "raw", digits = 0)

hist(trimmed_data$rt)
```




```r
range(trimmed_data$rt)
```

```
## [1]   245 11646
```

```r
hist(trimmed_data$pictureRT)
```




```
range(trimmed_data$pictureRT)
```

```
## [1]   245 11646
```

```
trimmed_data$condition <- trimmed_data$cond
trimmed_data$accuracy <- trimmed_data$acc
```

Convert to factor

```
trimmed_data$id <- droplevels(as.factor(trimmed_data$id))
```

Subset experiment and filler items

```
trimmed_data_esq <- subset(trimmed_data, experiment == "ES_Q")
trimmed_data_fc <- subset(trimmed_data, experiment == "FC")
trimmed_data_hom <- subset(trimmed_data, experiment == "Homogeneity") # additional experiment, included
trimmed_data_si <- subset(trimmed_data, experiment == "SI")
trimmed_data_upto <- subset(trimmed_data, experiment == "Up_to")
trimmed_data_fillers <- subset(trimmed_data, experiment == "Filler")
```

Exp1

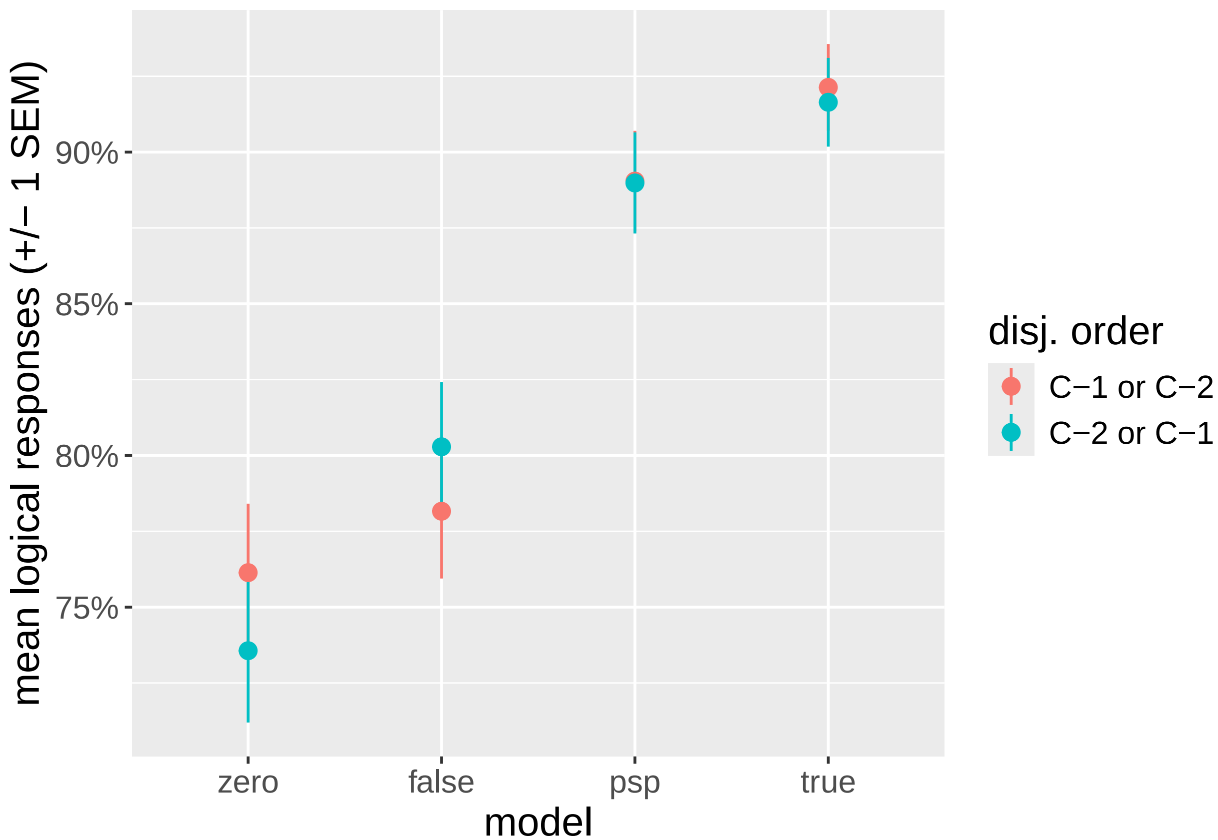
Name quantifiers, models, etc. and recombine data frames

```
###ESQ
trimmed_data_esq$quantifier = case_when(trimmed_data_esq$condition %in% c(1,2,3,4) ~ "less",
```

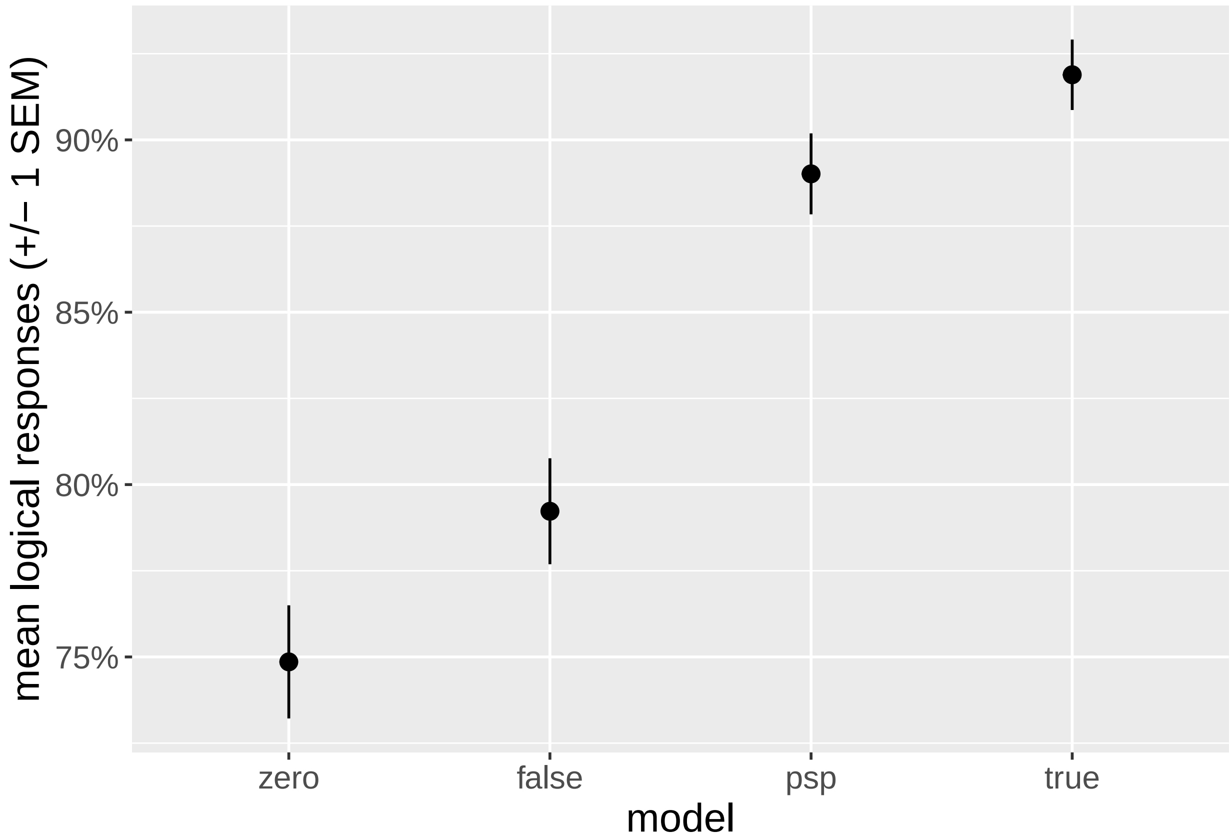
```
                                        trimmed_data_esq$condition %in% c(5,6,7,8) ~ "more",
                                        trimmed_data_esq$condition %in% c(9,10,11,12) ~ "every",
                                        trimmed_data_esq$condition %in% c(13,14,15,16) ~ "no")

trimmed_data_esq$model = case_when(trimmed_data_esq$condition %in% c(1,5,9,13) ~ "psp",
                                   trimmed_data_esq$condition %in% c(2,6,10,14) ~ "empty_set",
                                   trimmed_data_esq$condition %in% c(3,8,12) ~ "true",
                                   trimmed_data_esq$condition %in% c(4,7,11,15,16) ~ "false")

trimmed_data_esq$q_type = case_when(trimmed_data_esq$condition %in% 1:8 ~ "comparative",
                                    trimmed_data_esq$condition %in% 9:16 ~ "aristotelean")

trimmed_data_esq$mono = case_when(trimmed_data_esq$condition %in% c(1:4,13:16) ~ "down",
                                  trimmed_data_esq$condition %in% c(5:8,9:12) ~ "up")


###DIST
trimmed_data_fc$model = case_when(trimmed_data_fc$condition %in% c(1,5) ~ "psp",
                                  trimmed_data_fc$condition %in% c(2,6) ~ "empty_set",
                                  trimmed_data_fc$condition %in% c(3,7) ~ "true",
                                  trimmed_data_fc$condition %in% c(4,8) ~ "false")

trimmed_data_fc$disj_order = case_when(trimmed_data_fc$condition %in% 1:4 ~ "left",
                                       trimmed_data_fc$condition %in% 5:8 ~ "right")

##SI
trimmed_data_si$model = case_when(trimmed_data_si$condition %in% c(1) ~ "implicature",
                                  trimmed_data_si$condition %in% c(2) ~ "literal")

##Homogeneity
trimmed_data_hom$model = case_when(trimmed_data_hom$condition %in% c(1) ~ "psp",
                                   trimmed_data_hom$condition %in% c(4) ~ "true",
                                   trimmed_data_hom$condition %in% c(3) ~ "inhom",
                                   trimmed_data_hom$condition %in% c(2) ~ "false")
```

Generate plots for accuracy (always relative to logical responses)

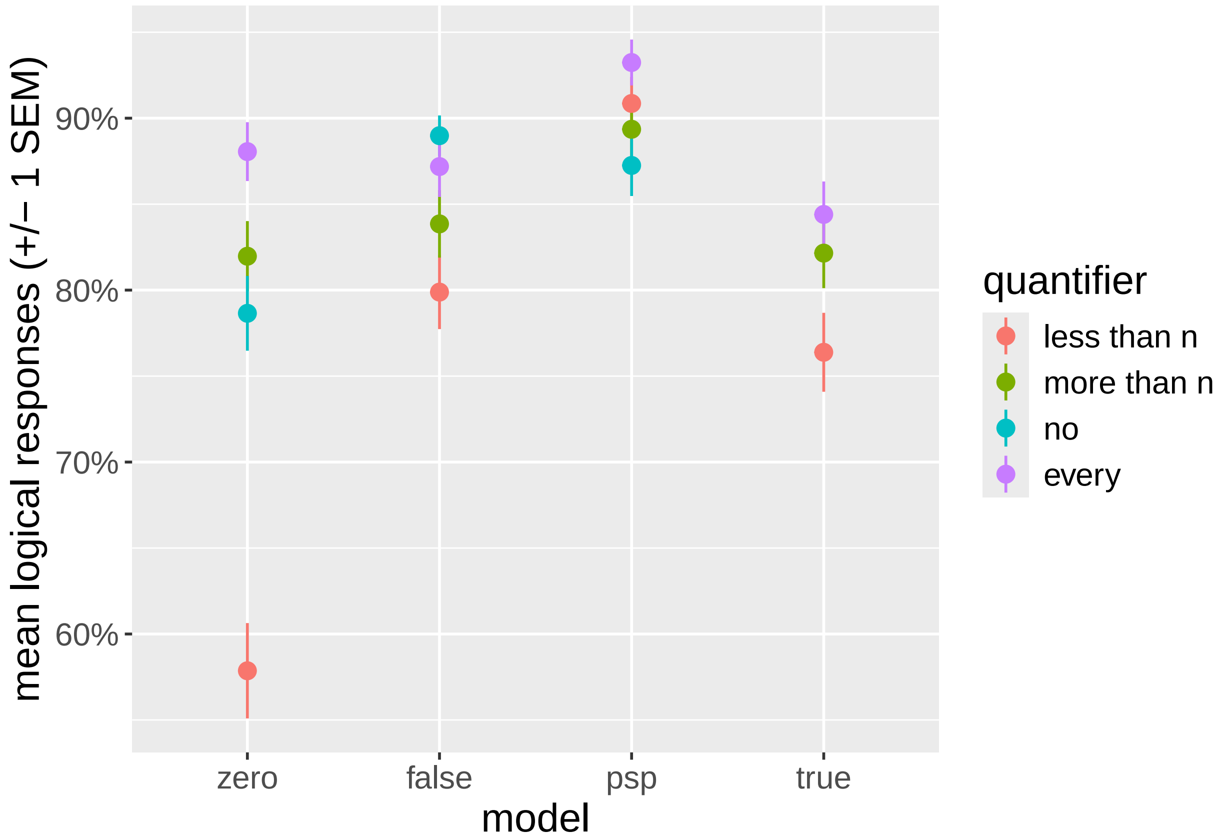
```
plot_fc_acc <- ggplot(trimmed_data_fc, aes(y=accuracy, x=model, color=disj_order)) +
  stat_summary(fun = mean,
               geom = "pointrange",
               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x))) +
  scale_y_continuous(labels = scales::percent) +
  theme(text = element_text(size = 15)) +
  ylab("mean logical responses (+/- 1 SEM)") +
  scale_x_discrete(breaks=c("empty_set","false", "psp","true"),
                   labels=c("zero","false", "psp","true")) +
  scale_colour_discrete(labels = c("C-1 or C-2", "C-2 or C-1")) +
  theme(text = element_text(size = 15))  + labs(colour = "disj. order")


plot_fc_acc
```



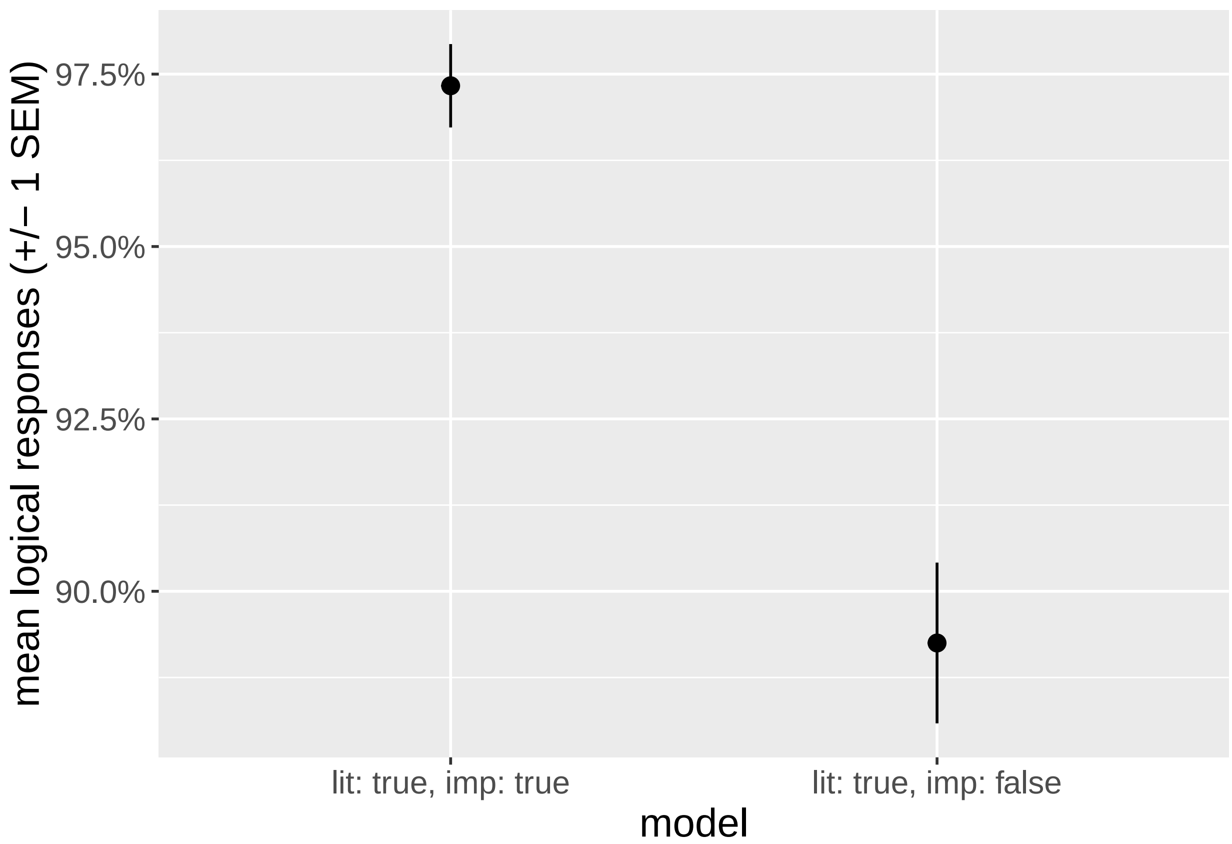

```r
ggsave(file.path(plots_dir,"plot_dist_acc.pdf"), width = 7, height = 7)
ggsave(file.path(plots_dir,"plot_dist_acc.png"), width = 7, height = 7)

plot_fc_acc_simp <- ggplot(trimmed_data_fc, aes(y=accuracy, x=model)) +
  stat_summary(fun = mean,
               geom = "pointrange",
               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x))) +
  scale_y_continuous(labels = scales::percent) +
  theme(text = element_text(size = 15)) +
  ylab("mean logical responses (+/- 1 SEM)") +
  scale_x_discrete(breaks=c("empty_set","false", "psp","true"),
                   labels=c("zero","false", "psp","true"))  +
  theme(text = element_text(size = 15))


plot_fc_acc_simp
```




```
ggsave(file.path(plots_dir,"plot_dist_acc_simp.pdf"), width = 7, height = 7)
ggsave(file.path(plots_dir,"plot_dist_acc_simp.png"), width = 7, height = 7)

trimmed_data_esq$quantifier <- factor(trimmed_data_esq$quantifier, levels = c("less", "more", "no", "eve

plot_esq_acc <- ggplot(trimmed_data_esq, aes(y=accuracy, x=model, colour = quantifier)) +
  stat_summary(fun = mean,
               geom = "pointrange",
               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x))) +
  scale_y_continuous(labels = scales::percent) +
  theme(text = element_text(size = 15)) +
  ylab("mean logical responses (+/- 1 SEM)") +
  scale_x_discrete(breaks=c("empty_set","false", "psp","true"),
                   labels=c("zero","false", "psp","true"))

plot_esq_acc
```




```
ggsave(file.path(plots_dir,"plot_esq_acc.pdf"), width = 7, height = 7)
ggsave(file.path(plots_dir,"plot_esq_acc.png"), width = 7, height = 7)

plot_si_acc <- ggplot(trimmed_data_si, aes(y=accuracy, x=model)) +
  stat_summary(fun = mean,
               geom = "pointrange",
               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x))) +
  scale_y_continuous(labels = scales::percent) +
  theme(text = element_text(size = 15)) +
  ylab("mean logical responses (+/- 1 SEM)") +
  scale_x_discrete(breaks=c("implicature","literal"),
                   labels=c("lit: true, imp: true","lit: true, imp: false"))

plot_si_acc
```

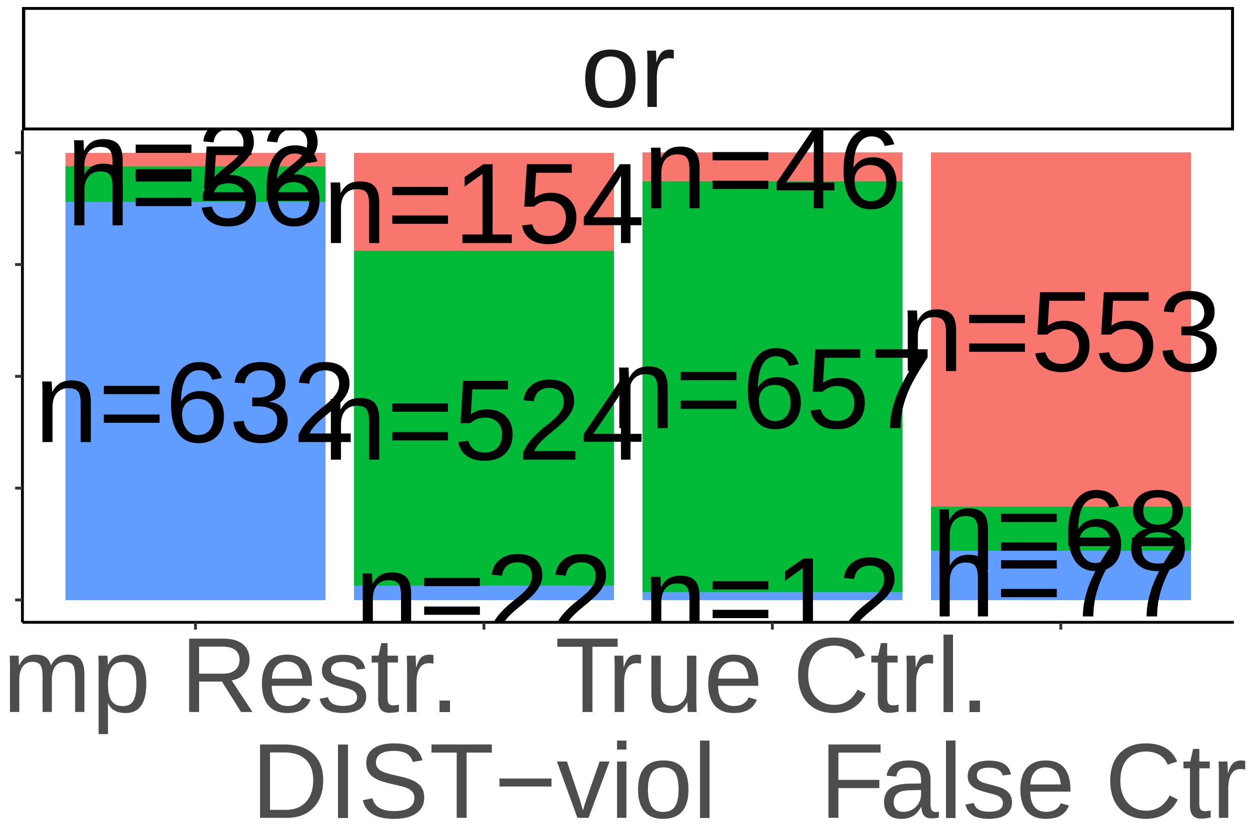



```r
ggsave(file.path(plots_dir,"plot_si_acc.pdf"), width = 4, height = 7)
ggsave(file.path(plots_dir,"plot_si_acc.png"), width = 4, height = 7)

#plot_hom_acc <- ggplot(trimmed_data_hom, aes(y=accuracy, x=model)) +
#  stat_summary(fun = mean,
#               geom = "pointrange",
#               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
#               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x))) +
#  scale_y_continuous(labels = scales::percent) +
#  theme(text = element_text(size = 15)) +
#  ylab("mean logical responses (+/- 1 SEM)")

#plot_hom_acc

#ggsave(file.path(plots_dir,"plot_hom_acc.pdf"), width = 7, height = 7)
#ggsave(file.path(plots_dir,"plot_hom_acc.png"), width = 7, height = 7)
```

Stacked bar charts showing all three responses

```r
# stacked bar chart for FC
trimmed_data_fc$given_response <- as.factor(trimmed_data_fc$given_response)

stacked_fc <- trimmed_data_fc %>%
  mutate(model = fct_relevel(model,
              "psp", "empty_set",  "true", "false")) %>%
```

```
  mutate(given_response = fct_relevel(given_response,
              "nein, stimmt nicht", "ja, stimmt",  "komische Frage")) %>%
  mutate(quantifier='or') %>%
  ggplot(aes(x = model, fill = given_response)) +
  geom_bar(position = "fill", stat = "count") + scale_y_continuous(labels = scales::percent) +
  scale_x_discrete(guide = guide_axis(n.dodge = 2),
                   breaks=c("psp","empty_set", "true","false"),
                   labels=c("Emp Restr.","DIST-viol", "True Ctrl.","False Ctrl.")) +

  geom_text(size=15,aes(label = paste0("n=", after_stat(count))), stat='count', position = position_fil
  theme_classic() +
  theme(text = element_text(size = 50)) +
  theme(legend.position = 'none')+
  theme(axis.title.y=element_blank(), axis.text.y=element_blank(), axis.title.x=element_blank())+
  scale_fill_discrete(labels = c("no, incorrect", "yes, correct", "weird question")) +
  labs(fill = "Response") +
  facet_grid(~quantifier)

stacked_fc
```

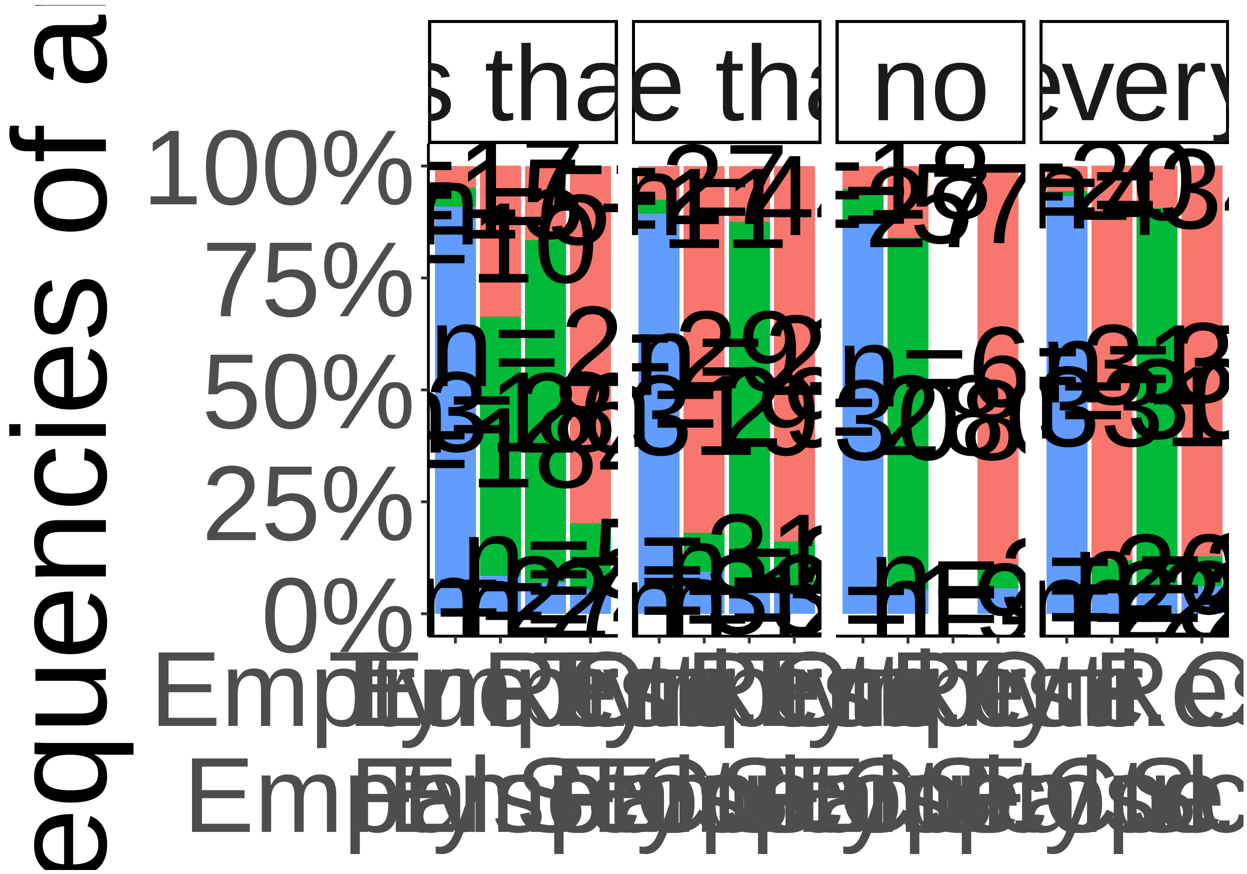



```
ggsave(file.path(plots_dir,"stacked_dist.pdf"), width = 8, height = 15)
ggsave(file.path(plots_dir,"stacked_dist.png"), width = 8, height = 15)


# stacked bar chart for Homogeneity
#trimmed_data_hom$given_response <- as.factor(trimmed_data_hom$given_response)
```

```r
#stacked_hom <- trimmed_data_hom %>%
#  mutate(given_response = fct_relevel(given_response,
#              "nein, stimmt nicht", "ja, stimmt",  "komische Frage")) %>%
#  ggplot(aes(x = model, fill = given_response)) +
#  geom_bar(position = "fill", stat = "count") + scale_y_continuous(labels = scales::percent) +
#  ylab("distributions of answers") +
#  geom_text(aes(label = paste0("n=", after_stat(count))), stat='count', position = position_fill(vjust
#  theme_classic()+
#  theme(text = element_text(size = 35)) +
#  scale_fill_discrete(labels = c("no, incorrect", "yes, correct", "weird question")) +
#  labs(fill = "Response")

#stacked_hom

#ggsave(file.path(plots_dir,"stacked_hom.pdf"), width = 12, height = 7)
#ggsave(file.path(plots_dir,"stacked_hom.png"), width = 12, height = 7)
```

##change figure width for facet grid and ESQ

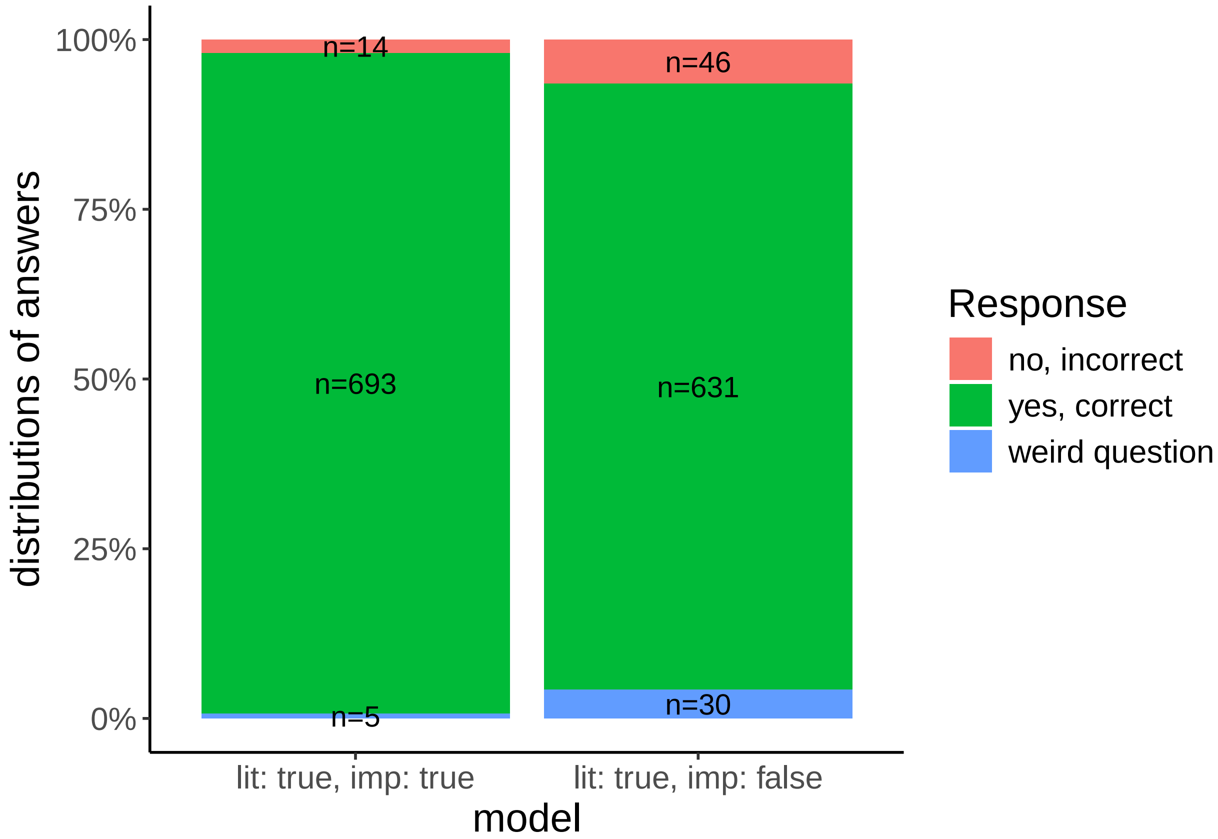
```r
# stacked bar chart for ESQ
trimmed_data_esq$given_response <- as.factor(trimmed_data_esq$given_response)

stacked_esq <- trimmed_data_esq %>%
  mutate(model = fct_relevel(model,
              "psp", "empty_set",  "true", "false")) %>%
  mutate(given_response = fct_relevel(given_response,
              "nein, stimmt nicht", "ja, stimmt",  "komische Frage"))  %>%
  ggplot(aes(x = model, fill = given_response)) +
  geom_bar(position = "fill", stat = "count") + scale_y_continuous(labels = scales::percent) + ylab("re
  geom_text(size=15,aes(label = paste0("n=", after_stat(count))), stat='count', position = position_fil
  theme_classic() +
  theme(text = element_text(size = 50)) +
  scale_x_discrete(guide = guide_axis(n.dodge = 2),
                   breaks=c("psp", "empty_set",  "true", "false"),
                   labels=c("Empty Restr.","Empty Scope", "True Ctrl.","False Ctrl."))  +
  scale_fill_discrete(labels = c("no, incorrect", "yes, correct", "odd question")) +
  theme(text = element_text(size = 50),axis.title.x=element_blank()) +
  theme(legend.position = 'none')+
  labs(fill = "Response") +
  facet_grid(~ quantifier)

stacked_esq
```




```
ggsave(file.path(plots_dir,"stacked_esq.pdf"), width = 36, height = 15)
ggsave(file.path(plots_dir,"stacked_esq.png"), width = 36, height = 15)
```

#adjust figure width for SI

```
# stacked bar chart for SI

trimmed_data_si$given_response <- as.factor(trimmed_data_si$given_response)

stacked_si <- trimmed_data_si %>%
  mutate(given_response = fct_relevel(given_response,
            "nein, stimmt nicht", "ja, stimmt",  "komische Frage")) %>%
  ggplot(aes(x = model, fill = given_response)) +
  geom_bar(position = "fill", stat = "count") + scale_y_continuous(labels = scales::percent) + ylab("di
  geom_text(aes(label = paste0("n=", after_stat(count))), stat='count', position = position_fill(vjust =
  scale_x_discrete(breaks=c("implicature","literal"),
                   labels=c("lit: true, imp: true","lit: true, imp: false"))  +
  theme_classic() +
  theme(text = element_text(size = 18)) +
  scale_fill_discrete(labels = c("no, incorrect", "yes, correct", "weird question")) +
  theme(text = element_text(size = 15)) +
  labs(fill = "Response")

stacked_si
```




```
ggsave(file.path(plots_dir,"stacked_si.pdf"))
```

```
## Saving 6.5 x 4.5 in image
```

```
ggsave(file.path(plots_dir,"stacked_si.png"))
```

```
## Saving 6.5 x 4.5 in image
```

Encode

Encode legit responses

```
###ESQ
trimmed_data_esq$legit_responses_one = case_when(trimmed_data_esq$condition %in% c(2,3,8,12,14)
                                            ~ case_when(tolower(trimmed_data_esq$given_response)
                                                        == tolower("ja, stimmt") ~ TRUE,
                                                        .default = FALSE),
                                            trimmed_data_esq$condition %in% c(1,5,9,13)
                                            ~ case_when(tolower(trimmed_data_esq$given_response)
                                                        == tolower("komische Frage") ~ TRUE,
                                                        .default = FALSE),
                                            .default=FALSE)

trimmed_data_esq$legit_responses_two = case_when(trimmed_data_esq$condition
                                                %in% c(2,4,6,7,10,11,14,16,15)
                                                ~ case_when(tolower(trimmed_data_esq$given_response)
                                                            == tolower("nein, stimmt nicht") ~ TRUE,
```

```
                                                                    .default = FALSE),
                                                      .default = FALSE)



###DIST
trimmed_data_fc$legit_responses_one = case_when(trimmed_data_fc$condition %in% c(2,3,6,7)
                                                ~ case_when(tolower(trimmed_data_fc$given_response)
                                                            == tolower("ja, stimmt") ~ TRUE,
                                                            .default = FALSE),
                                                trimmed_data_fc$condition %in% c(1,5)
                                                ~ case_when(tolower(trimmed_data_fc$given_response)
                                                            == tolower("komische Frage") ~ TRUE,
                                                            .default = FALSE),
                                                .default=FALSE)

trimmed_data_fc$legit_responses_two = case_when(trimmed_data_fc$condition
                                                    %in% c(2,6,4,8)
                                                    ~ case_when(tolower(trimmed_data_fc$given_response)
                                                                == tolower("nein, stimmt nicht") ~ TRUE,
                                                                .default = FALSE),
                                                    .default = FALSE)



###SI
trimmed_data_si$legit_responses_one = case_when(trimmed_data_si$condition %in% c(1,2)
                                                ~ case_when(tolower(trimmed_data_si$given_response)
                                                            == tolower("ja, stimmt") ~ TRUE,
                                                            .default = FALSE),
                                                .default=FALSE)

trimmed_data_si$legit_responses_two = case_when(trimmed_data_si$condition
                                                    %in% c(2)
                                                    ~ case_when(tolower(trimmed_data_si$given_response)
                                                                == tolower("nein, stimmt nicht") ~ TRUE,
                                                                .default = FALSE),
                                                    .default = FALSE)
```

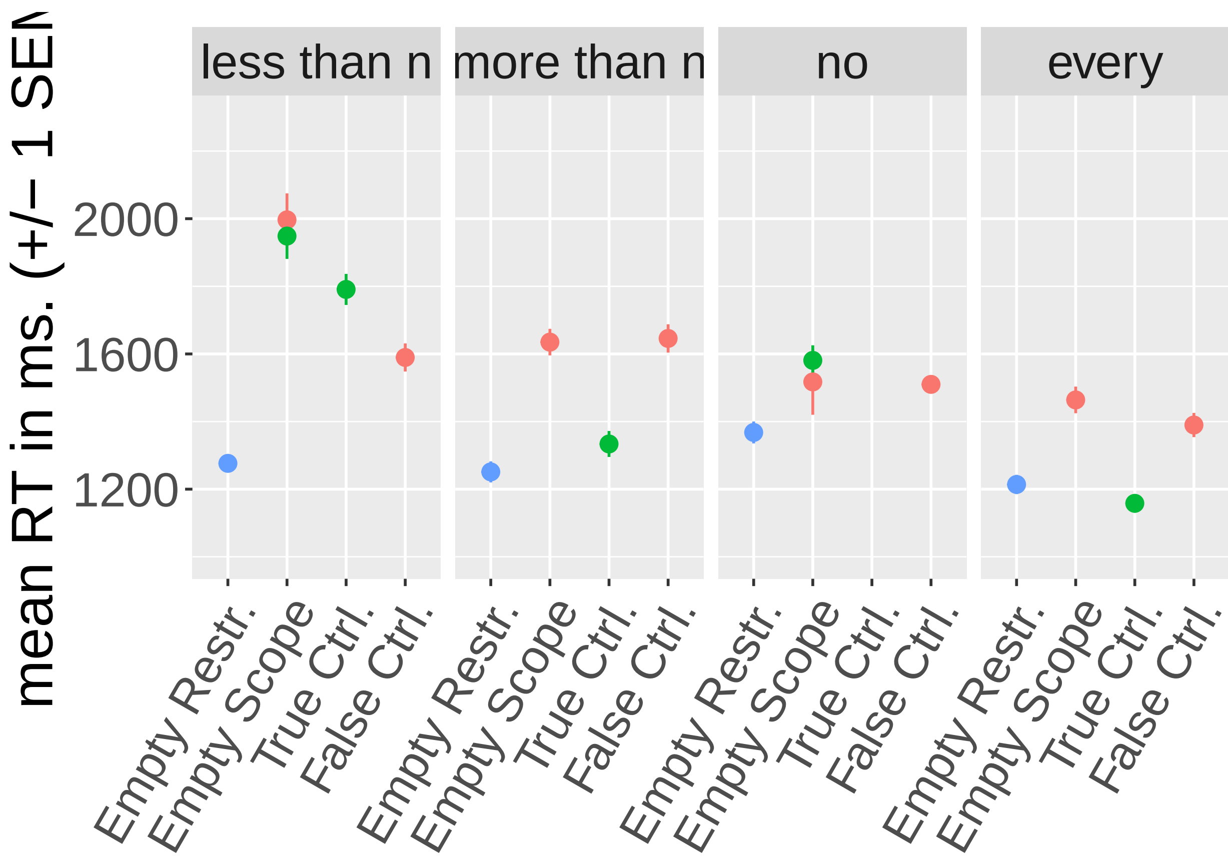
Subset to legit responses

```
###ESQ
trimmed_data_esq_legit <- subset(trimmed_data_esq, trimmed_data_esq$legit_responses_one | trimmed_data_

###DIST
trimmed_data_fc_legit <- subset(trimmed_data_fc, trimmed_data_fc$legit_responses_one | trimmed_data_fc$

###SI
trimmed_data_si_legit <- subset(trimmed_data_si, trimmed_data_si$legit_responses_one | trimmed_data_si$
```

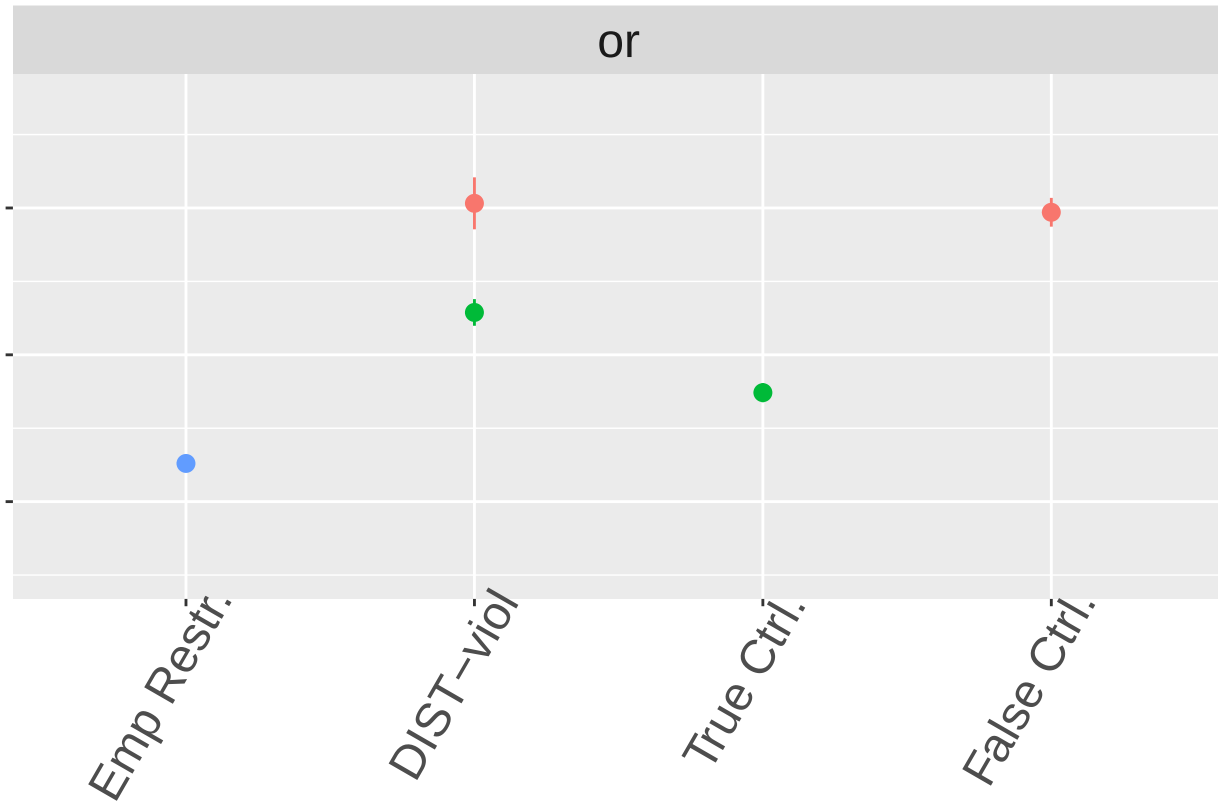
Plot RT of legit responses

```
###ESQ
plot_esq_rt_legit <- trimmed_data_esq_legit  %>%
```

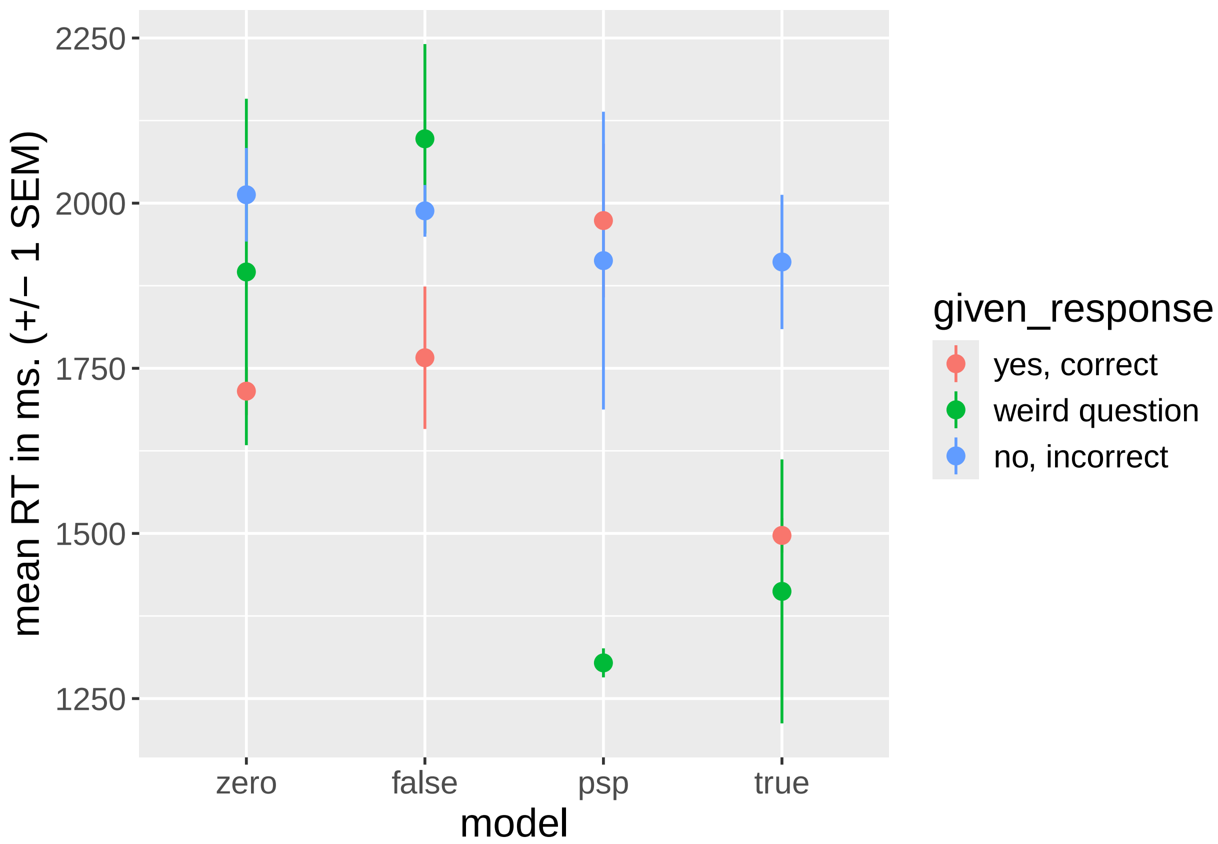
```
  mutate(model = fct_relevel(model,
              "psp", "empty_set",  "true", "false")) %>%
  mutate(given_response = fct_relevel(given_response,
              "nein, stimmt nicht", "ja, stimmt",  "komische Frage"))  %>%
  ggplot(aes(y=pictureRT, x=model, colour = given_response)) +
  stat_summary(fun = mean,
               geom = "pointrange",
               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x))) +
  #scale_y_continuous(labels = scales::percent) +
  ylab("mean RT in ms. (+/- 1 SEM)") +
   scale_x_discrete(breaks=c("psp", "empty_set",  "true", "false"),
                    labels=c("Empty Restr.","Empty Scope", "True Ctrl.","False Ctrl."))  +
  scale_colour_discrete(labels = c("no, incorrect", "yes, correct", "weird question")) +

  facet_grid(~ quantifier) +
  theme(text = element_text(size = 22), axis.title.x=element_blank(), axis.text.x = element_text(angle =
  theme(legend.position = 'none')+
  coord_cartesian(ylim = c(1000, 2300))+
  labs(colour = "Response")



plot_esq_rt_legit
```



```
ggsave(file.path(plots_dir,"plot_esq_rt_legit.pdf"), width = 12, height = 6)
ggsave(file.path(plots_dir,"plot_esq_rt_legit.png"), width = 12, height = 6)



###DIST
plot_fc_rt_legit <- trimmed_data_fc_legit  %>%
  mutate(quantifier='or') %>%
    mutate(given_response = fct_relevel(given_response,
              "nein, stimmt nicht", "ja, stimmt",  "komische Frage"))  %>%
  mutate(model = fct_relevel(model,
              "psp", "empty_set",  "true", "false")) %>%
  ggplot( aes(y=pictureRT, x=model, colour = given_response)) + stat_summary(fun = mean,
               geom = "pointrange",
               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x)))+
  #scale_y_continuous(labels = scales::percent) +
  ylab("mean RT in ms. (+/- 1 SEM)") +
 scale_x_discrete(breaks=c("psp", "empty_set",  "true", "false"),
                   labels=c("Emp Restr.","DIST-viol", "True Ctrl.","False Ctrl."))  +
  scale_colour_discrete(labels = c("no, incorrect", "yes, correct", "weird question")) +
       theme(legend.position = 'none')+
  theme(text = element_text(size = 22))  +
   facet_grid(~ quantifier) +
  theme(axis.title.y=element_blank(), axis.text.y=element_blank(),axis.title.x=element_blank(),axis.tex
  coord_cartesian(ylim = c(1000, 2300))+
  labs(colour = "Response")

plot_fc_rt_legit
```




```
#ggsave(file.path(plots_dir,"plot_fc_rt_legit.pdf"), width = 4, height = 5)
#ggsave(file.path(plots_dir,"plot_fc_rt_legit.png"), width = 4, height = 5)
ggsave(file.path(plots_dir,"plot_dist_rt_legit_wo.pdf"), width = 3, height = 6)
ggsave(file.path(plots_dir,"plot_dist_rt_legit_wo.png"), width = 3, height = 6)

plot_fc_rt <- ggplot(trimmed_data_fc, aes(y=pictureRT, x=model, colour = given_response)) + stat_summar
               geom = "pointrange",
               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x)))+
  #scale_y_continuous(labels = scales::percent) +
  ylab("mean RT in ms. (+/- 1 SEM)") +
    scale_x_discrete(breaks=c("empty_set","false", "psp","true"),
                   labels=c("zero","false", "psp","true")) +
  scale_colour_discrete(labels = c("yes, correct", "weird question", "no, incorrect")) +

  theme(text = element_text(size = 15))

plot_fc_rt
```




```
ggsave(file.path(plots_dir,"plot_dist_rt.pdf"))
```

```
## Saving 6.5 x 4.5 in image
```

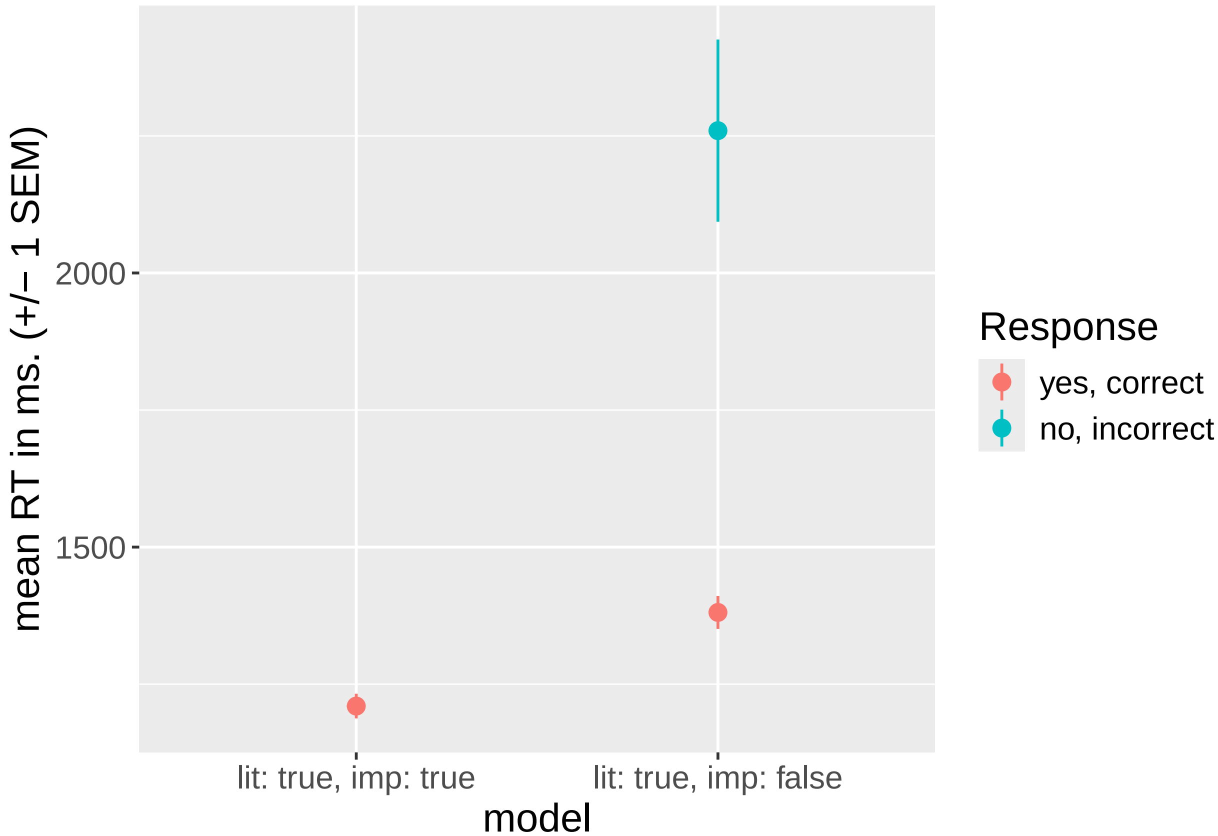
```
ggsave(file.path(plots_dir,"plot_dist_rt.png"))
```

```
## Saving 6.5 x 4.5 in image
```

```
plot_si_rt_legit <- trimmed_data_si_legit  %>%
  mutate(model = fct_relevel(model,
            "psp", "empty_set",  "true", "false")) %>% ggplot(aes(y=pictureRT, x=model, colour = giver
             geom = "pointrange",
             fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
             fun.min = function(x) mean(x) - sd(x)/sqrt(length(x)))+
  #scale_y_continuous(labels = scales::percent) +
  ylab("mean RT in ms. (+/- 1 SEM)") +
  #scale_x_discrete(guide = guide_axis(n.dodge = 2)) +
  scale_x_discrete(breaks=c("implicature", "literal"),
                 labels=c("lit: true, imp: true", "lit: true, imp: false")) +
  scale_colour_discrete(labels = c("yes, correct", "no, incorrect")) +
  theme(text = element_text(size = 15)) +
  labs(colour = "Response")
```

```
## Warning: 4 unknown levels in `f`: psp, empty_set, true, and false
```

```r
plot_si_rt_legit
```




```r
ggsave(file.path(plots_dir,"plot_si_rt_legit.pdf"))
```

```
## Saving 6.5 x 4.5 in image
```

```r
ggsave(file.path(plots_dir,"plot_si_rt_legit.png"))
```

```
## Saving 6.5 x 4.5 in image
```

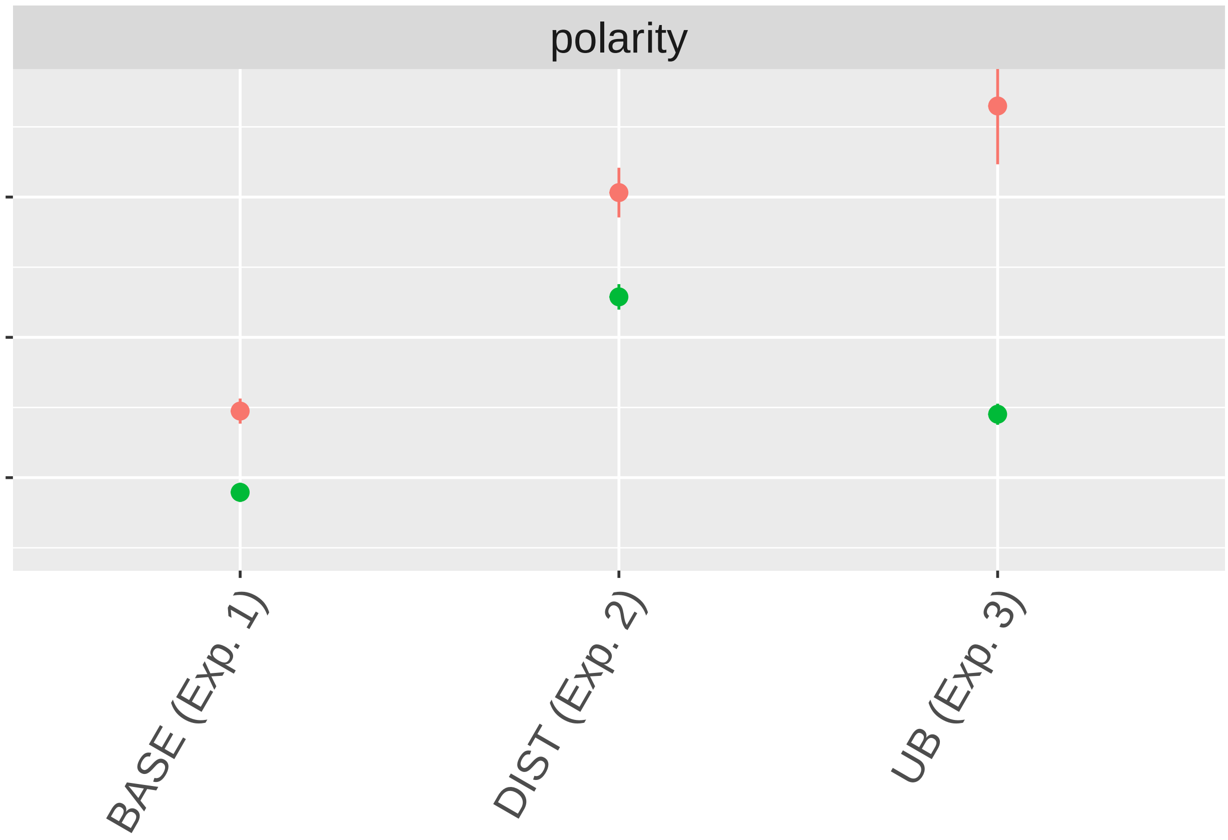
#Merge data for comparison between response polarity effects of distributivity vs scalar implicature inferences

Comparison within UE Aristotelean quantifiers. Include control conditions for *every* from ESQ subexperiment as baseline comparisons.

```r
###ESQ
trimmed_data_esq$selection_responses_one = case_when(trimmed_data_esq$condition %in% c(12)
                                             ~ case_when(tolower(trimmed_data_esq$given_response)
                                                         == tolower("ja, stimmt") ~ TRUE,
                                                         .default = FALSE),
                                             .default=FALSE)

trimmed_data_esq$selection_responses_two = case_when(trimmed_data_esq$condition
                                                 %in% c(11)
                                                 ~ case_when(tolower(trimmed_data_esq$given_response)
                                                             == tolower("nein, stimmt nicht") ~ TRUE,
                                                             .default = FALSE),
                                                 .default = FALSE)
```

```
###FC
trimmed_data_fc$selection_responses_one = case_when(trimmed_data_fc$condition %in% c(2,6)
                                            ~ case_when(tolower(trimmed_data_fc$given_response)
                                                        == tolower("ja, stimmt") ~ TRUE,
                                                        .default = FALSE),
                                            .default=FALSE)

trimmed_data_fc$selection_responses_two = case_when(trimmed_data_fc$condition
                                                %in% c(2,6)
                                                ~ case_when(tolower(trimmed_data_fc$given_response)
                                                            == tolower("nein, stimmt nicht") ~ TRUE,
                                                            .default = FALSE),
                                                .default = FALSE)

###SI
trimmed_data_si$selection_responses_one = case_when(trimmed_data_si$condition %in% c(2)
                                            ~ case_when(tolower(trimmed_data_si$given_response)
                                                        == tolower("ja, stimmt") ~ TRUE,
                                                        .default = FALSE),
                                            .default=FALSE)

trimmed_data_si$selection_responses_two = case_when(trimmed_data_si$condition
                                                %in% c(2)
                                                ~ case_when(tolower(trimmed_data_si$given_response)
                                                            == tolower("nein, stimmt nicht") ~ TRUE,
                                                            .default = FALSE),
                                                .default = FALSE)

###ESQ
trimmed_data_esq_selection <- subset(trimmed_data_esq, trimmed_data_esq$selection_responses_one | trimme

###FC
trimmed_data_fc_selection <- subset(trimmed_data_fc, trimmed_data_fc$selection_responses_one | trimmed_

###SI
trimmed_data_si_selection <- subset(trimmed_data_si, trimmed_data_si$selection_responses_one | trimmed_

###merge dataframes

df_list <- list(trimmed_data_esq_selection, trimmed_data_fc_selection, trimmed_data_si_selection)
trimmed_data_selection <- Reduce(function(x, y) merge(x, y, all=TRUE), df_list, accumulate=FALSE)

###plot RTs of all conditions in merged dataframe

plot_rt_selection <- trimmed_data_selection  %>%
   mutate(quantifier='polarity') %>%
  mutate(given_response = fct_relevel(given_response,
            "nein, stimmt nicht", "ja, stimmt",  "komische Frage"))  %>%
  ggplot(aes(y=pictureRT, x=experiment, colour = given_response)) +
  stat_summary(fun = mean,
               geom = "pointrange",
               fun.max = function(x) mean(x) + sd(x)/sqrt(length(x)),
               fun.min = function(x) mean(x) - sd(x)/sqrt(length(x))) +
```

```
  #scale_y_continuous(labels = scales::percent) +
  ylab("mean RT in ms. (+/- 1 SEM)") +
  xlab("compared models") +
  scale_x_discrete(breaks=c("ES_Q","FC", "SI"),
                   labels=c("BASE (Exp. 1)","DIST (Exp. 2)", "UB (Exp. 3)")) +
  scale_colour_discrete(type= c("#F8766D", "#00BA38", "#619CFF"),labels = c("no, false","yes, true",  "
  labs(colour = "Response") +
  theme(text = element_text(size = 20))  +
   facet_grid(~ quantifier) +
        theme(legend.position = 'none')+
  theme(axis.title.y=element_blank(), axis.text.y=element_blank(),axis.title.x=element_blank(), axis.te
  coord_cartesian(ylim = c(1000, 2300))

plot_rt_selection
```




```
#ggsave(file.path(plots_dir,"plot_dist_vs_si_answer_polarity_effect.pdf"), width = 5, height = 5)
#ggsave(file.path(plots_dir,"plot_dist_vs_si_answer_polarity_effect.png"), width = 5, height = 5)
ggsave(file.path(plots_dir,"plot_dist_vs_si_answer_polarity_effect_wo.pdf"), width = 4, height = 6)
ggsave(file.path(plots_dir,"plot_dist_vs_si_answer_polarity_effect_wo.png"), width = 4, height = 6)
```

Save all

```
save(trimmed_data, trimmed_data_esq, trimmed_data_fc, trimmed_data_si, trimmed_data_hom, trimmed_data_u
```